

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belgaum-590018



Mini Project Report

on

“Krishimitra - AI ML Based Decision Support System For Sustainable Farming”

Submitted in partial fulfillment of the requirements for the
Second Semester of the Bachelor of Engineering Degree, towards the completion of the
Mini Project under the **Interdisciplinary Project**,
Department of Basic Sciences.

by

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CERTIFICATE

This is to certify that the File Structures mini project entitled “**AI ML Based Decision Support System For Sustainable Farming**” has been successfully carried out by **Vinmithashri (1CR25CS211), Vishal M (1CR25CS212), Waseem Raza (1CR25AD142), and Y Roopa (1CR25AD143)** are bonafide students of **CMR Institute of Technology**.

The project is submitted in partial fulfillment of the requirements for the First Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the **Interdisciplinary Project, Department of Basic Sciences**.

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

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Abstract

This project presents an AI-ML based Smart Farmer Decision Support Algorithm designed to assist farmers in making efficient, data-driven agricultural decisions. The system follows a structured multi-step process, beginning with language selection and input collection, including soil parameters (N, P, K, pH, moisture, and soil type), weather data (temperature, humidity, and rainfall), and location details.

The collected data is validated and analyzed using machine learning techniques to assess soil health, detect nutrient deficiencies, and determine fertility levels.

Based on this analysis, the system recommends suitable crops, fertilizers, and irrigation strategies, ensuring optimal resource utilization and improved productivity. It also integrates real-time weather forecasting through APIs and provides water management suggestions to enhance sustainability.

Additionally, the system includes market price analysis to help farmers choose the most profitable crops.

All recommendations are presented in the user's preferred language for better accessibility. The data is stored for future analysis and continuous model improvement.

Overall, this solution promotes smart farming by improving productivity, sustainability, and farmer income through informed decision-making.